Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Practical	Tut	Theory	Practical	Tut	Total
PEC7011	Natural Language Processing	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract/ Oral	Total
		Internal Assessment			End SEM Exam	Exam duration (in hrs)			
		Test 1	Test 2	Avg.					
PEC7011	Natural Language Processing	20	20	20	80	3	-	-	100

Course Objectives: The Course aims to

1. To understand natural language processing and to learn how to apply basic algorithms in this field.
2. To get acquainted with the basic concepts and algorithmic description of the main language levels: morphology, syntax, semantics, and pragmatics.
3. To design and implement various language models and POS tagging techniques.
4. To design and learn NLP applications such as Information Extraction, Question answering.
5. To design and implement applications based on natural language processing.

Course Outcomes: Learners will be able to

1) To have a broad understanding of the field of natural language processing
2) To design language model for word level analysis for text processing
3) To design various POS tagging techniques
4) To design, implement and test algorithms for semantic analysis
5) To develop basic understanding of Pragmatics and to formulate the discourse segmentation and anaphora resolution
6) To apply NLP techniques to design real world NLP applications

Prerequisite: Some prior knowledge about Java programming, Basics of SQL, Data mining and machine learning methods would be beneficial.

DETAILED SYLLABUS: total six module for each subject (13 Weeks)

Sr. No.	Module	Detailed Content	Hours	CO Mapping
I	**Introduction**	1.1 Origin & History of NLP, 1.2 The need of NLP, 1.3 Generic NLP System, 1.4 Levels of NLP, 1.5 Knowledge in Language Processing, 1.6 Ambiguity in Natural Language, Challenges of NLP, 1.7 Applications of NLP.	**04**	**CO1** **CO6**
II	**Word Level Analysis**	2.1 Tokenization, Stemming, Segmentation, Lemmatization, Edit , Distance, Collocations, 2.2 Finite Automata, Finite State Transducers (FST), Porter Stemmer, 2.3 Morphological Analysis, Derivational and Reflectional Morphology, 2.4 Regular expression with types. 2.5 N –Grams, Unigrams/Bigrams 2.6 Language Models, Corpora, 2.7 Computing the Probability of Word Sequence 2.8 Training and Testing.	**09**	**CO1** **CO2**
III	**Syntax analysis**	3.1 Part-Of-Speech Tagging (POS) - Open and Closed Words. Tag Set for English (Penn Treebank), 3.2 Rule Based POS Tagging, Transformation Based Tagging, 3.3 Stochastic POS Tagging and Issues –Multiple Tags & Words, Unknown Words. 3.4 Introduction to CFG, 3.5 Hidden Markov Model (HMM), 3.6 Maximum Entropy And Conditional Random Field (CRF).	**09**	**CO2** **CO3**
IV	**Semantic Analysis**	4.1 Introduction, meaning representation 4.2 Lexical Semantics 4.3 Corpus study 4.4 Study of Various language dictionaries like WordNet, Babelnet 4.5 Relations among lexemes & their senses – Homonymy, Polysemy, Synonymy, Hyponymy	**09**	**CO3** **CO4**

		4.6 Semantic Ambiguity 4.7 Word Sense Disambiguation (WSD) 4.8 Knowledge based approach (Lesk's Algorithm), Supervised (Naïve Bayes, Decision List) 4.9 Introduction to Semi-supervised method (Yarowsky), Unsupervised (Hyperlex)		
V	**Pragmatic & Discourse Processing**	5.1 Discourse: Reference Resolution, Reference Phenomena, Syntactic & Semantic constraint on coherence; 5.2 Anaphora Resolution using Hobbs and Cantering Algorithm	**06**	**CO4** **CO5**
VI	**Applications (preferably for Indian regional languages)**	6.1 Machine Translation, Information Retrieval, Question Answers System, Categorization, Summarization, Sentiment Analysis, Named Entity Recognition. 6.2 Linguistic Modeling – 6.3 Neurolinguistics Models- 6.4 Psycholinguistic Models – 6.5 Functional Models of Language – Research Linguistic Models- Common Features of Modern Models of Language.	**05**	**CO1** **CO6**
		Total No. of Lectures:	**42**	

Text Books:

1. Daniel Jurafsky, James H. and Martin, Speech and Language Processing, Second Edition, Prentice Hall, 2008.
2. Christopher D.Manning and HinrichSchutze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.

References:

1. Siddiqui and Tiwary U.S., Natural Language Processing and Information Retrieval, Oxford University Press, 2008.
2. Daniel M Bikel and ImedZitouni — Multilingual natural language processing applications: from theory to practice, IBM Press, 2013.
3. Nitin Indurkhya and Fred J. Damerau, —Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.

Online References:

Sr. No.	Website Name
1.	https://onlinecourses.nptel.ac.in/noc21_cs102/preview
2.	https://onlinecourses.nptel.ac.in/noc20_cs87/preview
3.	https://nptel.ac.in/courses/106105158

Assessment:

Internal Assessment (IA) for 20 marks:

- ✓ IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test.

Question paper format:

- Question Paper will comprise of a total of **six questions each carrying 20 marks Q.1** will be **compulsory** and should **cover maximum contents of the syllabus.**
- From the **remaining five questions, attempt any three. The Remaining five questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules).
- A total of **Four questions** need to be answered.